



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG246
Course Name: Thermodynamics II
Semester/ Year: Spring 2023-2024
Instructor: Assoc. Prof. Dr. Devrim Aydin
Assistant: Mr. Mehdi Ahmad/Mr. Bijan Rashidinia
Assessment: Lab
Due Date: May 2024 at 19:00 hrs.

LAB NAME: Air conditioning

STUDENT(S) NO.: 23704019

NAME(S), SURNAME(S): TARIK TARIQ

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Introduction	6	8	10		
Theory	6	8	10		
Apparatus	6	8	10		
Procedure	6	8	10		
Calculations & Results	6	8	30		
Discussion & Conclusion	6	8	30		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	Understand the operating principles of gas powered thermodynamic cycles	X						
2	Understand the operating principles of vapour and combined power cycles	X						
3	Understand the operating principles of different refrigeration cycles (reversed carnot, vapour compression, absorption, advanced refrigeration cycles) and analyze their performances based on thermodynamic principles.	X						
4	Analyze the properties and p-v-t behavior of gas mixtures.	X						
5	Perform thermodynamic analysis on different processes (heating, cooling, humidification and dehumidification) by using thermodynamic relations and	X						
6	Understand the properties of different fuels and write thermodynamic	X						
7	Analyze the thermodynamic performance of wet cooling tower experimentally	X					X	
8	Analyze the thermodynamic performance of an HVAC process experimentally	X					X	
	Weight of Student Outcomes	H					M	

Instructions:

- A. Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.
- B. Submission:** Upload your report as a pdf file to the related section in LMS within the designated time. The file uploaded should be named as **[MENG246-S2022-2023-Lab2-Student ID-Student Name]**. No late submissions will be accommodated.

ABSTRACT

This study investigates the performance of the Argemsan K-212 Recycle Heat Recovery Air Handling Unit, an advanced system designed for energy-efficient air conditioning. The unit's ability to regulate temperature and humidity while recovering thermal energy from exhaust air is analyzed through experiments simulating heating and cooling processes. Inlet and outlet conditions are measured, and mass flow rates are calculated to assess the system's effectiveness. Results demonstrate the unit's capacity to increase outlet temperature while decreasing relative humidity during heating, and vice versa during cooling, highlighting its energy recovery capabilities. These findings underscore the importance of the Argemsan K-212 in sustainable building management, providing valuable insights for HVAC professionals towards optimizing system performance and contributing to energy efficiency goals.

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1. INTRODUCTION

Air conditioning is crucial for maintaining indoor comfort by regulating temperature, humidity, and air quality in various settings. With a growing emphasis on energy efficiency, advanced systems like the Argemsan K-212 Recycle Heat Recovery Air Handling Unit have emerged to meet these needs.

The Argemsan K-212 enhances energy efficiency by capturing and reusing thermal energy from exhaust air, reducing the energy required to condition incoming fresh air. This not only saves costs but also lowers greenhouse gas emissions, supporting global sustainability efforts.

As a training set, the Argemsan K-212 provides hands-on experience with heat recovery mechanisms, helping trainees understand HVAC system operations and the importance of energy recovery. By studying this unit, future HVAC professionals can learn to optimize system performance and contribute to sustainable building management.

2. THEORY

In simple heating and cooling processes, the temperature of the air stream is controlled, resulting in a higher outlet temperature for heating and a lower outlet temperature for cooling. However, since there is no addition or removal of moisture during these processes, the specific humidity remains constant, meaning ω_{inlet} equals ω_{outlet} . Conversely, because relative humidity is influenced by the dry air temperature, it will change during the process, meaning ϕ_{inlet} is not equal to ϕ_{outlet} .

The outlet air temperature is influenced by the air mass flow rate, which will be demonstrated and analyzed during the experiment. The mass flow rate of the air flowing through the unit can be calculated using the following equation:

$$\dot{m}_{\text{air}} = \rho V A_c \quad (1)$$

where $\dot{m}_{\text{air}}$ is the air mass flow rate, ρ is the air density, V is the air velocity, and A_c is the cross-sectional area of the flow.

3. APPARATUS

The Argemsan K-212 Recycle Heat Recovery Air Handling Unit is an advanced training apparatus designed to demonstrate the principles and operations of energy-efficient air conditioning systems. The unit incorporates several key components that simulate real-world HVAC processes, making it an invaluable educational tool.

Key Components and Features

1. Display Panel:

- The display panel provides real-time data and control options for the system. It includes a digital screen that shows various operational parameters and allows for adjustments to be made easily.

2. Control Interface:

- The control interface at the top features several switches and dials, including the main switch, emergency button, leakage relay, insurance, and fan speed control. These controls help in managing the system's operations and safety.

3. Low Pressure Manometer:

- A low pressure manometer is included to measure and display the pressure within the system, ensuring it operates within the desired parameters.

4. Heat Recovery Wheel:

- The central component of the unit, the heat recovery wheel, facilitates the transfer of thermal energy between the exhaust and supply air streams. It is positioned at the core of the apparatus and is essential for demonstrating heat recovery processes.

5. Air Filters:

- The unit is equipped with high-efficiency air filters (T3-H3, T2-H2, T4-H4) to remove particulates from both incoming and outgoing air streams, ensuring clean air and maintaining system efficiency.

6. Heaters:

- The preheater and last heater are used to condition the air to the desired temperature. They can be controlled via the switches on the control interface.

7. Humidification Pump:

- The humidification pump controls the moisture content in the air, though in basic heating and cooling processes, no additional moisture is added or removed.

8. Fans and Blowers:

- The fans and blowers, located at various points (e.g., T1-H1), ensure consistent airflow through the system, which is crucial for effective heat exchange and ventilation.

9. Sensors and Measurement Points:

- The unit includes various sensors and labeled measurement points (e.g., T1-H1, T2-H2, T3-H3, T4-H4, T5-H5, T6-H6, T7-H7) to monitor temperature and humidity at different stages of the process.

10. Airflow Pathways:

- The pathways for air movement are clearly marked with arrows, illustrating the direction of airflow through the different sections of the unit. This helps in understanding the flow dynamics and the heat recovery process

The Argemsan K-212 operates by utilizing the heat recovery wheel to transfer heat between the exhaust and supply air streams. During the heating process, the wheel captures heat from the warm exhaust air and transfers it to the cooler incoming air, increasing its temperature. In the cooling process, the wheel helps to transfer heat from the warm incoming air to the cooler exhaust air, reducing the cooling load on the system.

As no moisture is added or removed during these processes, the specific humidity remains constant. However, the relative humidity will change due to the temperature variations between the inlet and outlet air streams.



Figure 1 Argemsan Apparatus

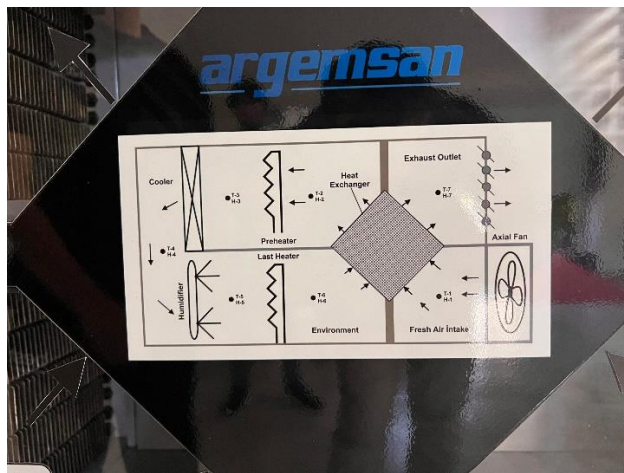


Figure 2 Argemsan apparatus description

Description of Argemsan K-212 Recycle Heat Recovery Air Handling Unit

The Argemsan K-212 Recycle Heat Recovery Air Handling Unit is designed to demonstrate energy-efficient HVAC processes. It features:

- Heat Exchanger: The central component for transferring heat between the exhaust and supply air streams.
- Cooler and Heaters: Includes a cooler, preheater, and last heater to condition the air as needed.
- Humidifier: Adds moisture to the air when necessary.
- Axial Fan: Ensures consistent airflow through the system.
- Air Filters and Sensors: Positioned at various points (T1-H1, T2-H2, T3-H3, etc.) to monitor temperature and humidity, ensuring clean and efficient operation.

The labeled diagram shows the airflow pathways and the locations of these components, illustrating how the unit manages and recycles heat to maintain optimal indoor air conditions.

4. PROCEDURE

To begin the experiment, turn on the fuses and main switch, then start the fan on the control panel and adjust its speed to an average of 1.2 m/s using a handheld anemometer. Next, activate the preheater (or chiller for cooling processes) on the control panel and wait for the temperatures to stabilize. Once stable, record the inlet and outlet temperature and relative humidity values for the preheater (or chiller). Repeat the experiment by adjusting and setting different air speeds on the anemometer and varying the fan speed accordingly.

5. CALCULATIONS AND RESULTS

For Simple Heating; velocity (V) = 1.2 m/s

$$\text{Area (Ac)} = 0.1095 \text{ m}^2$$

Table 1 simple heating data

DATA	INLET CONDITION	OUTLET CONDITION
T(°C)	26.1	34.8
Φ(%)	52.7	36.5
TO FIND FROM PSYCHROMETRIC CHART		
ω(Kg.H ₂ O/Kg.dry air)	0.0111	0.0127
h(KJ/Kg.dry air)	55.0	68
v(m ³)	0.863	0.89

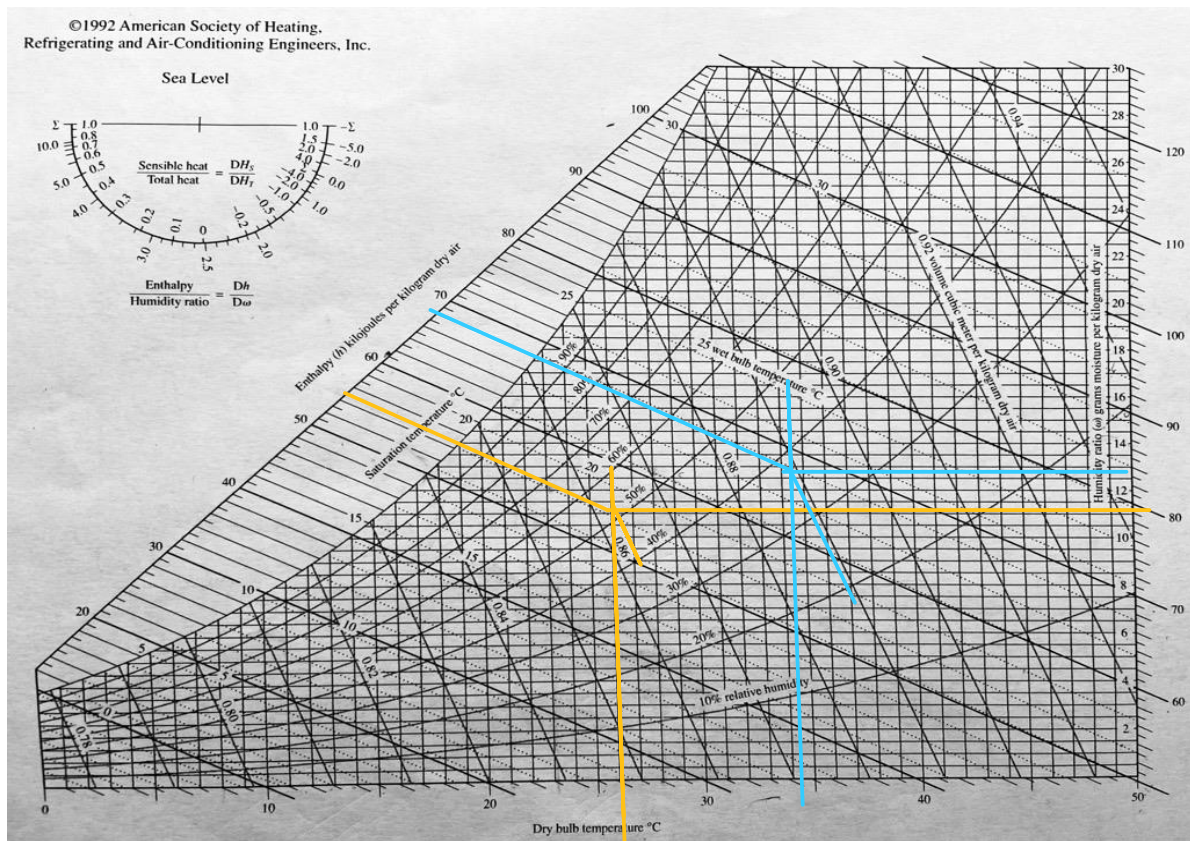


Figure 3

Using the equation (1) which states $\dot{m}_{air} = \rho V A c$

$$V = 1.2 \text{ m/s}$$

$$A c = 0.1095 \text{ m}^2$$

$$\rho = 1/v$$

$$\dot{m}_{air} = \frac{A c \times V}{v} \text{ (kg/s)}$$

Table 2 inlet and outlet data

INLET	OUTLET
-------	--------

$\dot{m}_{air} = \frac{Ac \times V}{v} (kg/s)$ $\dot{m}_{air} = \frac{0.1095 \times 1.2}{0.863}$ $\dot{m}_{air(inlet-heating)} = 0.1523 \text{ Kg/s}$	$\dot{m}_{air} = \frac{Ac \times V}{v} (kg/s)$ $\dot{m}_{air} = \frac{0.1095 \times 1.2}{0.89}$ $\dot{m}_{air(outlet-heating)} = 0.1476 \text{ Kg/s}$
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For Simple Cooling; velocity (V) = 1.2 m/s

$$\text{Area (Ac)} = 0.1095 \text{ m}^2$$

Table 3 simple cooling data

DATA	INLET CONDITION	OUTLET CONDITION
T(°C)	26.7	21.2
Φ(%)	52.6	56.6
TO FIND FROM PSYCHROMETRIC CHART		
ω(Kg.H ₂ O/Kg.dry air)	0.0115	0.00898
h(KJ/Kg.dry air)	56.4	44
v(m ³)	0.865	0.846

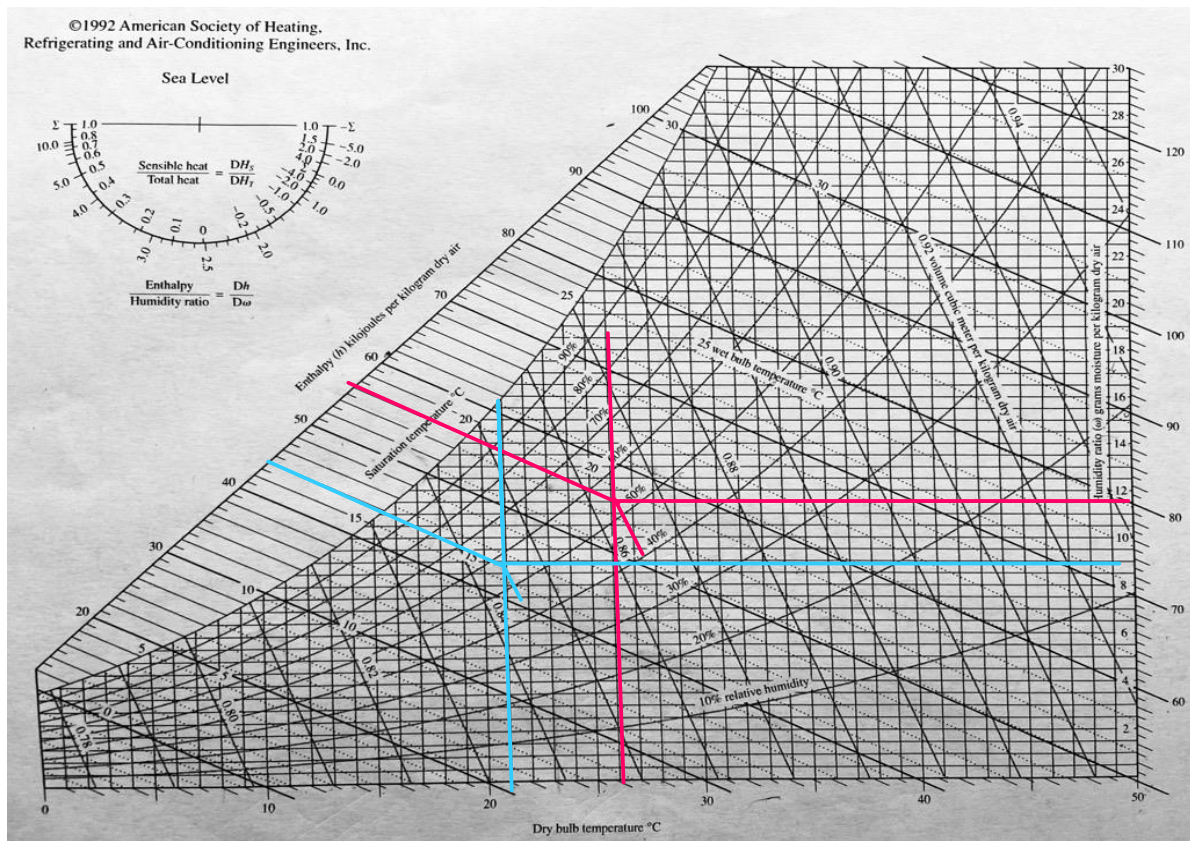


Figure 4

Using the equation (1) which states $\dot{m}_{air} = \rho V A_c$

$$V = 1.2 \text{ m/s}$$

$$A_c = 0.1095 \text{ m}^2$$

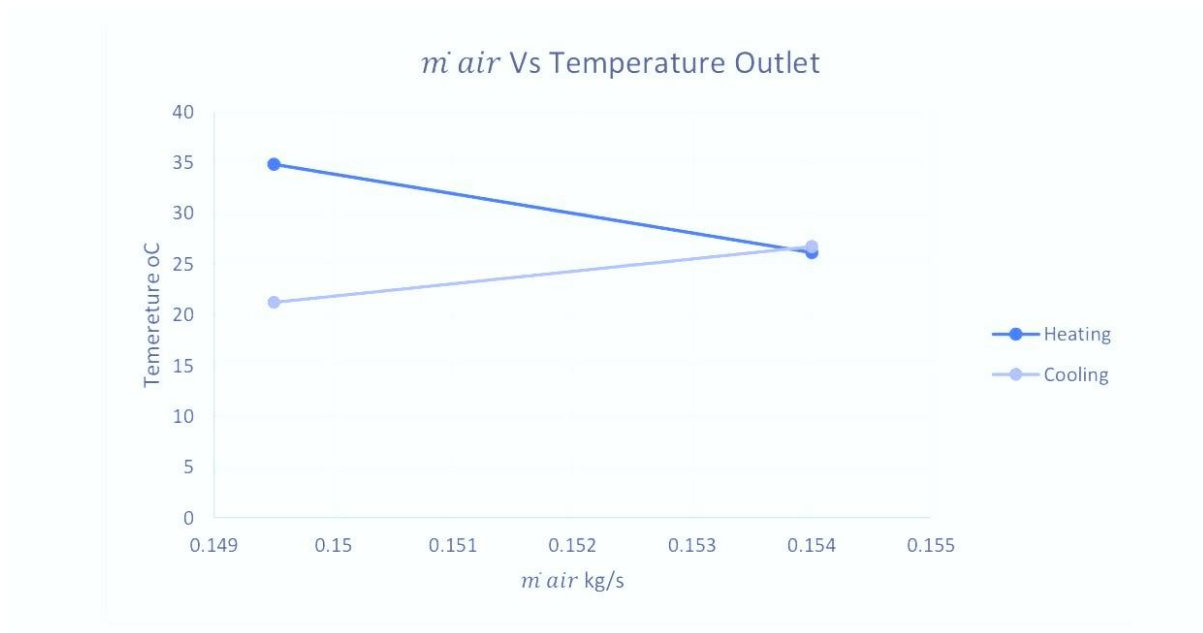
$$\rho = 1/\nu$$

$$\dot{m}_{air} = \frac{A_c \times V}{\nu} \text{ (kg/s)}$$

Table 4 inlet and outlet mass flowrates

INLET	OUTLET
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$\dot{m}_{air} = \frac{Ac \times V}{v} \text{ (kg/s)}$ $\dot{m}_{air} = \frac{0.1095 \times 1.2}{0.865}$ $\dot{m}_{air}(\text{inlet-cooling}) = \mathbf{0.1519 \text{ Kg/s}}$	$\dot{m}_{air} = \frac{Ac \times V}{v} \text{ (kg/s)}$ $\dot{m}_{air} = \frac{0.1095 \times 1.2}{0.846}$ $\dot{m}_{air}(\text{outlet-cooling}) = \mathbf{0.1553 \text{ Kg/s}}$
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graph 1 mass flowrate vs outlet temperature graph

6. DISCUSSION AND CONCLUSION

The experiment conducted with the Argemsan K-212 Recycle Heat Recovery Air Handling Unit showcases its efficacy in both heating and cooling processes. By measuring inlet and outlet conditions for temperature and relative humidity and calculating mass flow rates, we discerned significant insights into its performance. The unit effectively regulates air properties, demonstrating its ability to increase outlet temperature while decreasing relative humidity during heating and vice versa during cooling. These outcomes affirm the unit's efficiency in energy recovery and air conditioning, crucial for sustainable building management. The slight variations in mass flow rates between inlet and outlet indicate consistent airflow maintenance despite temperature and humidity adjustments, underscoring the system's reliability. These findings are invaluable for HVAC professionals, offering practical insights into system design, operation, and optimization towards energy efficiency and sustainability goals.

7. REFERENCES

(n.d.). Argemsan Eğitim Teknolojileri – Teknik Eğitime Atılan
İmza.... https://www.argemsan.com/Content/images/products/k-212-heat-recovery-type-air-conditioning-training-set_48871440.pdf